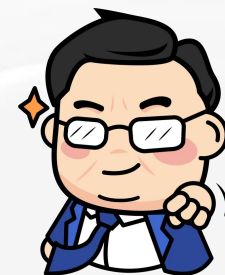


26高数强化 (15)

15	多元微分学的概念及举例（重极限、连续、偏导数及全微分）	P156-P165
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下 p165-178

主讲 武忠祥 教授



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你就慢了



@武忠祥老师 @考研可爱因子



第五章 多元函数微分学

对照：
相同
不同

✓ 第一节 重极限 连续 偏导数 全微分

第二节 偏导数与全微分的计算

第三节 极值与最值 ⊗

第一节 重极限 连续 偏导数 全微分

本节内容要点

一. 考试内容要点精讲

(一) 重极限

(二) 连续

(三) 偏导数

(四) 全微分

(五) 连续、可导、可微的关系



还不关注，
你就慢了



二. 常考题型方法与技巧

讨论连续性、可导性、可微性

一. 考试内容要点精讲

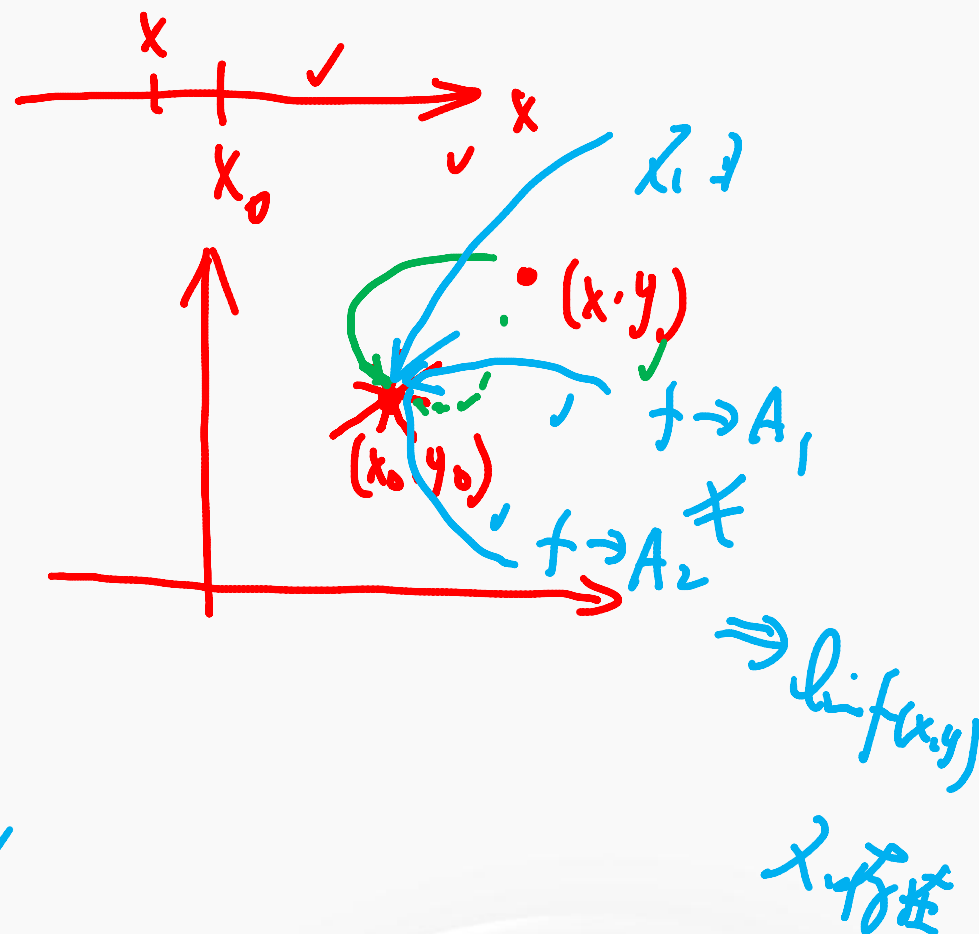
26武忠祥考研

(一) 重极限

$$\lim_{x \rightarrow x_0} f(x) = A$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = A$$

存在



注 1) $(x,y) \rightarrow (x_0,y_0)$ 是以“任意方式”

2) (1) 局部有界性 ✓

(2) 保号性 ✓

(3) 有理运算 ✓

(4) 极限与无穷小的关系 ✓

(5) 夹逼性 ✓

【例1】求下列极限

$$1) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$$

Handwritten notes: 3次 0, 2次 0, 2.3, $\rightarrow \infty$, $\rightarrow 0$

$$2) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{|x| + |y|}$$

Handwritten notes: 2次 0, 1次 0, 0

$$3) \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2 \sin(xy)}{x^2 + y^4}$$

Handwritten notes: 2, 4, 2, 4

Handwritten: $f(x) \rightarrow 0 \Leftrightarrow |f(x)| \rightarrow 0$

Handwritten: $2ab \leq a^2 + b^2$

Handwritten: ① $|f(x,y)|$, ② 夹逼

【解】1) $0 \leq \frac{x^2 y}{x^2 + y^2} \leq |y| \rightarrow 0$

$$2) 0 \leq \frac{x^2 + y^2}{|x| + |y|} = \frac{|x||x|}{|x| + |y|} + \frac{|y||y|}{|x| + |y|} \leq |x| + |y| \rightarrow 0$$

则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{|x| + |y|} = 0$

3. 方法1 由于 $\left| \frac{xy^2}{x^2 + y^4} \right| \leq \frac{1}{2}$, 即为有界量, 而

$\lim_{(x,y) \rightarrow (0,0)} \sin xy = 0$. 即为无穷小量, 则原式 $= 0$.

方法2 $0 \leq \frac{xy^2 \sin xy}{x^2 + y^4} \leq \frac{1}{2} |\sin xy| \rightarrow 0$

方法3 $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2 \sin(xy)}{x^2 + y^4} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{x^2 + y^4} = 0$

常用方法

1. 利用极限性质 (四则运算法则, 夹逼原理)
2. 消去分母中极限为零的因子 (有理化, 等价无穷小代换)
3. 利用无穷小量与有界变量之积为无穷小量.

【例2】证明下列重极限不存在

1) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$; ~~2次~~ ~~2次~~

【证】 1) $\lim_{\substack{y=kx \\ x \rightarrow 0}} \frac{xy}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{kx^2}{x^2 + k^2 x^2} = \frac{k}{1+k^2}$

则重极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2 + y^2}$ 不存在.

2) $\lim_{\substack{y=kx \\ x \rightarrow 0}} \frac{xy^2}{x^2 + y^4} = \lim_{x \rightarrow 0} \frac{k^2 x^3}{x^2 + k^4 x^4} = \lim_{x \rightarrow 0} \frac{k^2 x}{1 + k^4 x^2} = 0$ *

$\lim_{\substack{x=y^2 \\ y \rightarrow 0}} \frac{xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0} \frac{y^4}{y^4 + y^4} = \frac{1}{2} \neq 0$

常用方法

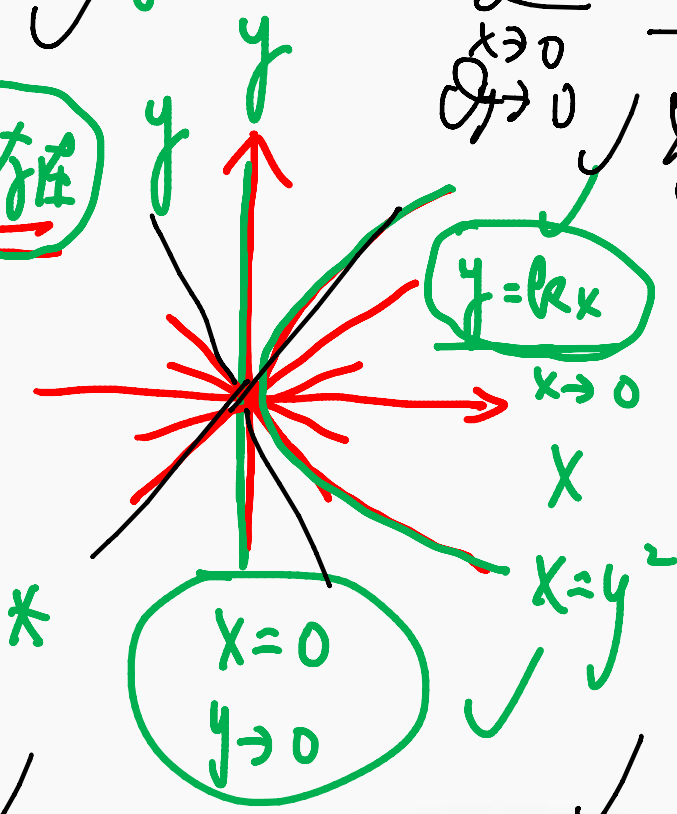
沿两种不同路径极限不同 (通常可取过点 (x_0, y_0) 的直线)

$\lim_{r \rightarrow 0} \frac{r^3 \cos \theta \sin^2 \theta}{r^2 \cos^2 \theta + r^4 \sin^2 \theta} = \lim_{r \rightarrow 0} \frac{r \cos \theta \sin^2 \theta}{\cos^2 \theta + r^2 \sin^2 \theta} = 0$

2) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$

$x=y^2$

不存在



$y = x^2$

$\lim_{x \rightarrow 0} \frac{x^2 y}{x^4 + y^2}$

$\lim_{r \rightarrow 0} f(r \cos \theta, r \sin \theta) = A \not\Rightarrow \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = A$?



(二) 连续

1) 定义 $\lim_{(x,y) \rightarrow (x_0,y_0)} \underline{f(x,y)} = \underline{f(x_0,y_0)}$

2) 性质

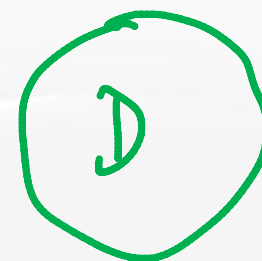
(1) 多元连续函数的和、差、积、商（分母不为零）及复合仍为连续函数。

(2) 多元基本初等函数在其定义域内连续；初等函数在其定义区域内连续。

(3) 有界闭区域上连续函数的性质

(a) 有界性：若 $f(x,y)$ 在有界闭区域 D 上连续，则 $f(x,y)$ 在 D 上有界。

多元 - 元 ① 定义
① 定义
② 间断 *
③ 性质
③ 性质



(b) **最值性**: 若 $f(x, y)$ 在 有界闭区域 D 上连续, 则 $f(x, y)$ 在 D 上必有 最大值和最小值.

(c) **介值性**: 若 $f(x, y)$ 在 有界闭区域 D 上连续, 则 $f(x, y)$ 在 D 上可取到介于 最小值 与 最大值 之间的任何值. ✓

【例3】判断函数 $f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}} & (x, y) \neq (0, 0) \\ a & (x, y) = (0, 0) \end{cases}$ 的连续性.

【解】因为 $0 \leq \frac{|xy|}{\sqrt{x^2 + y^2}} \leq |y|$, 则 $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0$.

若 $a = 0$, $f(x, y)$ 处处连续; 若 $a \neq 0$, $f(x, y)$ 除点 $(0, 0)$

外处处连续.

(3) 偏导数

1) 定义

$$\begin{aligned} \varphi(x) &= f(x, y_0) \\ \varphi'(x_0) &= \lim_{\Delta x \rightarrow 0} \frac{\varphi(x_0 + \Delta x) - \varphi(x_0)}{\Delta x} \\ f_x(x_0, y_0) &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \frac{d}{dx} f(x, y_0) \Big|_{x=x_0} \\ f_y(x_0, y_0) &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} = \frac{d}{dy} f(x_0, y) \Big|_{y=y_0} \end{aligned}$$

“先代后求”

【例4】设 $f(x, y) = x + 2y + (y-1)\arcsin \frac{x}{1+xy}$, 求 $f_x(0, 1), f_y(0, 1)$.

【解】

$$\begin{aligned} f_x(0, 1) &= \frac{d}{dx} f(x, 1) \Big|_{x=0} = \frac{d}{dx} (x + 2) \Big|_{x=0} = 1 \\ f_y(0, 1) &= \frac{d}{dy} f(0, y) \Big|_{y=1} = \frac{d}{dy} (2y) \Big|_{y=1} = 2; \end{aligned}$$

① $f_x(x, y)$
② $x=0, y=1$ 代入

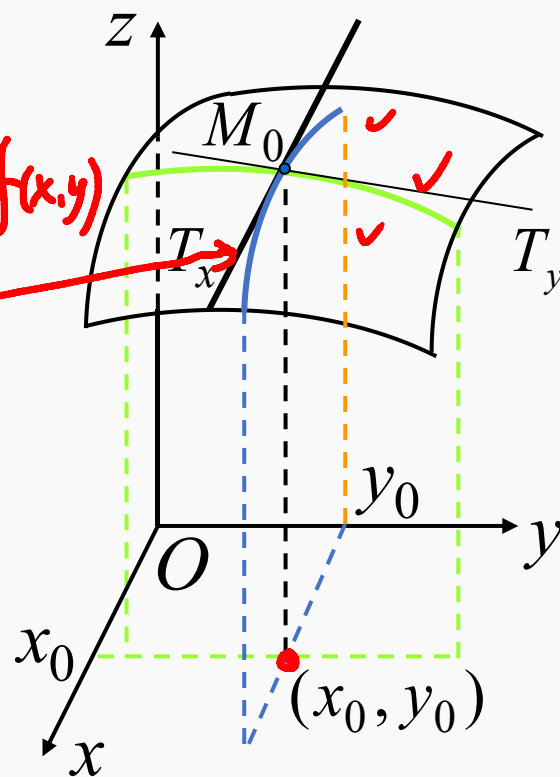
2) 几何意义

$z = f(x, y)$ 曲面

$z = f(x, y_0)$ 曲线

$f_x(x_0, y_0)$

$f_y(x_0, y)$



3) 高阶偏导数 设 $z = f(x, y)$

$$\checkmark \quad \checkmark$$

$$\underline{f_x(x, y)} \cdot \underline{f_y(x, y)}$$

$$\underline{\frac{\partial^2 z}{\partial x^2}} = f''_{xx}(x, y) = \underline{\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right)},$$

$$\frac{\partial^2 z}{\partial x \partial y} = f''_{xy}(x, y) = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right),$$

$$\frac{\partial^2 z}{\partial y \partial x} = f''_{yx}(x, y) = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right),$$

$$\frac{\partial^2 z}{\partial y^2} = f''_{yy}(x, y) = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right),$$

定理 如果函数 $z = f(x, y)$ 的两个二阶混合偏导数 $\underline{f''_{xy}(x, y)}$

及 $\underline{f''_{yx}(x, y)}$ 在区域 D 内连续, 则在区域 D 内恒有

$$f''_{xy}(x, y) = f''_{yx}(x, y)$$



(四) 全微分

1) 定义: 若

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = A\Delta x + B\Delta y + o(\rho)$$

$\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$
 $x \rightarrow x_0, y \rightarrow y_0$

【注】以下4条等价

$$\Delta \left\{ \begin{array}{l} (1) \Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = A\Delta x + B\Delta y + o(\rho); \\ (2) \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{[f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)] - [A\Delta x + B\Delta y]}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = 0; \\ (3) \Delta z = f(x, y) - f(x_0, y_0) = A(x - x_0) + B(y - y_0) + o(\rho); \\ (4) \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} \frac{[f(x, y) - f(x_0, y_0)] - [A(x - x_0) + B(y - y_0)]}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0. \end{array} \right.$$

$$A=1, B=0$$

$$= \Delta x + o(\rho) = o(\rho)$$

$$\Delta x + 2\Delta y = o(\rho)$$

2) 判定: $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = \underbrace{f_x \Delta x + f_y \Delta y}_{A\Delta x + B\Delta y} + o(\rho)$

$$A = f_x = 0$$

$$\exists A B \quad B = f_y = 0$$



(1) 必要条件 $f_x(x_0, y_0)$ 与 $f_y(x_0, y_0)$ 都存在;

(2) 充分条件 $f_x(x, y)$ 和 $f_y(x, y)$ 在 (x_0, y_0) 连续;

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f_x(x, y) = f_x(x_0, y_0)$$

* (3) 用定义判定

✓ a) $f_x(x_0, y_0)$ 与 $f_y(x_0, y_0)$ 是否都存在?

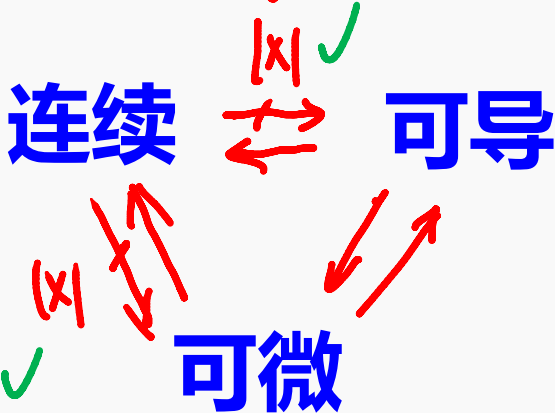
✓ b) $\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{\Delta z - [f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y]}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}$ 是否为零?

8:10

3) 计算: 若 $f(x, y)$ 可微, 则 $dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

(五) 连续、可导、可微的关系

一元函数



$$f(x,y) = \begin{cases} (x^2+y^2) \sin \frac{1}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (0,0) \end{cases}$$

多元函数

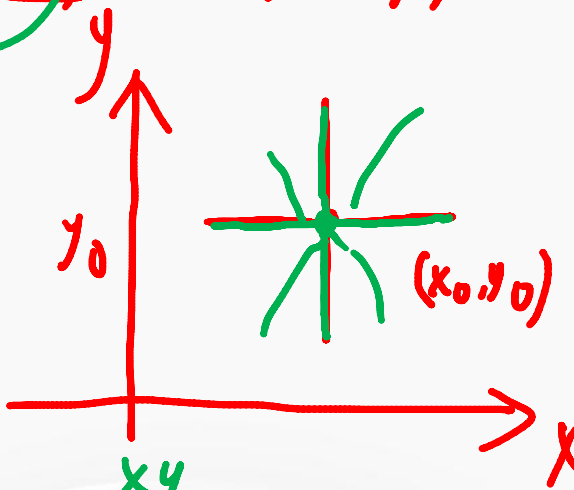


$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (0,0) \end{cases}$$

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (0,0) \end{cases}$$

$$f'_x(x_0, y_0) \quad f(x_0, y_0)$$

$$f'_y(x_0, y_0) \quad f(x_0, y)$$



$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (0,0) \end{cases}$$



✓ ✓ ✓
题型一 讨论连续性、可导性、可微性 *

【例1】 设 $f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$, 则 $f(x, y)$ 在 $(0, 0)$

- X A) 不连续;
 X B) 连续但不可导;
 ✓ C) 可导但不可微;
 D) 可微.

【解】 $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y}{x^2 + y^2} = 0 = f(0, 0)$

是线函数

①

$$f_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0$$

$$f_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0 - 0}{\Delta y} = 0$$

$$f(x, 0) = \begin{cases} 0, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

②

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{[f(\Delta x, \Delta y) - f(0, 0)] - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y]}{[(\Delta x)^2 + (\Delta y)^2]^{\frac{3}{2}}}$$

$\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$
 $\Delta y = k\Delta x$
 不存在
 3次

【例2】考虑二元函数下面四条性质

① $f(x, y)$ 在点 (x_0, y_0) 处连续; ✓

② $f(x, y)$ 在点 (x_0, y_0) 处两个偏导数连续; ✗

③ $f(x, y)$ 在点 (x_0, y_0) 处可微; ✓

④ $f(x, y)$ 在点 (x_0, y_0) 处两个偏导数都存在. 则

(A) ③ [✓] $\Rightarrow$ ① [✗] $\Rightarrow$ ④.

(B) ③ [✗] $\Rightarrow$ ② [✓] $\Rightarrow$ ①;

(C) ③ [✓] $\Rightarrow$ ④ [✗] $\Rightarrow$ ①;

(D) ② [✓] $\Rightarrow$ ③ [✓] $\Rightarrow$ ①;

【例3】二元函数 $f(x, y)$ 在点 $(0, 0)$ 处可微的一个充分条件是

✗ (A) $\lim_{(x, y) \rightarrow (0, 0)} [f(x, y) - f(0, 0)] = 0$ 错误.

✗ (B) $\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0$, 且 $\lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = 0$; 不对

✓ (C) $\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y) - f(0, 0)}{\sqrt{x^2 + y^2}} = 0$; ✓

✗ (D) $\lim_{x \rightarrow 0} [f_x(x, 0) - f_x(0, 0)] = 0$, 且 $\lim_{y \rightarrow 0} [f_y(0, y) - f_y(0, 0)] = 0$. ✓

✓
 $\lim_{x \rightarrow 0} f_x(x, y) = f_x(0, 0)$
 $\lim_{x \rightarrow 0} f_x(x, 0) = f_x(0, 0)$

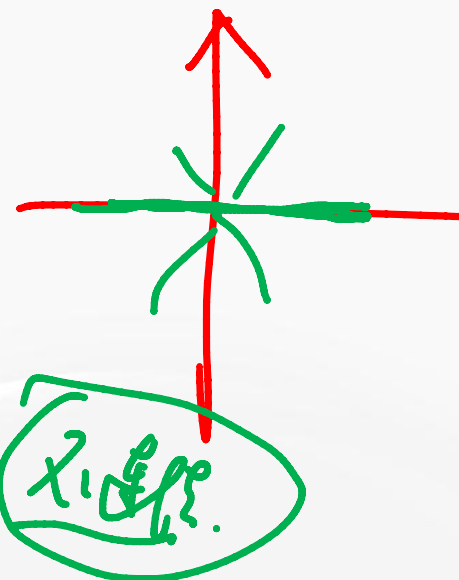
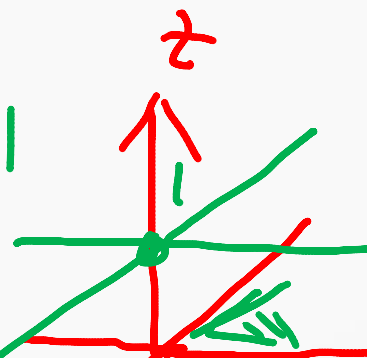
【解1】排除法 (A) (B) 显然不正确,

D 令 $f(x, y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$ ✓

$\lim_{x \rightarrow 0} [f_x(x, 0) - f_x(0, 0)] = 0$, 且 $\lim_{y \rightarrow 0} [f_y(0, y) - f_y(0, 0)] = 0$

但 $f(x, y)$ 在 $(0, 0)$ 点不连续, 因此不可微, (D) 不正确.

$f(x, 0) = 1$



$y=0$ $k \rightarrow 0$

【解2】直接法

由

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{\sqrt{x^2 + y^2}} = 0 \quad \text{知}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{\sqrt{x^2}} &= \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{|x|} \\ &= \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} \cdot \frac{x}{|x|} = 0 \end{aligned}$$

$$f_x(0,0) = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = 0 \quad \checkmark$$

同理 $f_y(0,0) = 0$. $\checkmark$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0) - [f_x(0,0)x + f_y(0,0)y]}{\sqrt{x^2 + y^2}} \\ = \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{\sqrt{x^2 + y^2}} = 0 \end{aligned}$$

则 $f(x,y)$ 在 $(0,0)$ 点处可微, 故 应选 (C)。

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$$\textcircled{1} \quad \underline{\underline{f_x(0,0) = 0.}}$$

 $\textcircled{2}$ 

【解3】直接法

由

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - f(0,0)}{\sqrt{x^2 + y^2}} = 0$$

A, B 0

$$f(x,y) - f(0,0) = Ax + By + o(\rho)$$

$x \rightarrow x_0$
 $y \rightarrow y_0$

$$f(x,y) - f(x_0,y_0)$$

$$f(x,y) - f(0,0) = o(\rho)$$

$\exists A, B$

$$\lim_{x \rightarrow 0, y \rightarrow 0} \frac{(x+y) \leq \frac{1}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}} = 0$$

$$f(x,y) = \begin{cases} (x^2+y^2) \leq \frac{1}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} \frac{[f(x,y) - f(x_0,y_0)] - [A(x-x_0) + B(y-y_0)]}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} = 0$$



【例4】 如果函数 $f(x,y)$ 在 $(0,0)$ 处连续, 那么下列

命题正确的是

命题正确的是

(A) 若极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y)}{|x| + |y|}$ 存在, 则 $f(x, y)$ 在 $(0, 0)$ 处可微.

✓ (B) 若极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y)}{x^2 + y^2}$ 存在, 则 $f(x, y)$ 在 $(0, 0)$ 处可微. ✓

(C) 若 $f(x, y)$ 在 $(0, 0)$ 处可微, 则极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y)}{|x| + |y|}$ 存在.

~~(D)~~ 若 $f(x, y)$ 在 $(0, 0)$ 处可微, 则极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x, y)}{x^2 + y^2}$ 存在. ∞

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x,y) - f(0,0)}{x^2 + y^2} \quad \checkmark$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{f(x,y) - f(0,0)}{\sqrt{x^2 + y^2}}$$



【例5】 设连续函数 $z = f(x, y)$ 满足

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} = 0, \text{ 则 } dz|_{(0,1)} = \underline{\quad\quad\quad}.$$

【解1】 由 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} = 0$ 得, $\underline{\underline{2dx - dy}}$

$f(0,1) = 1$, 且 $\lim_{x \rightarrow 0} \frac{f(x, y) - f(0,1)}{x} = 2$, $\lim_{y \rightarrow 1} \frac{f(x, y) - f(0,1)}{y-1} = 1$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{[f(x, y) - f(0,1)] - [2x + (y-1)]}{\sqrt{x^2 + (y-1)^2}} = 0$$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} \frac{[f(x, y) - f(x_0, y_0)] - [A(x - x_0) + B(y - y_0)]}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0.$$

$$df = A dx + B dy$$

【例5】 设连续函数 $z = f(x, y)$ 满足

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{f(x, y) - 2x + y - 2}{\sqrt{x^2 + (y-1)^2}} = 0, \text{ 则 } dz|_{(0,1)} = \underline{\quad 2dx - dy \quad}.$$

【解2】

$$f(x, y) - 2x + y - 2 = 0$$

$$\rightarrow f(x, y) = 2x - y + 2$$

$$f_x = 2, \quad f_y = -1$$

【例6】设 $f(x, y) = |x - y| \varphi(x, y)$, 其中 $\varphi(x, y)$ 在点 $(0, 0)$ 的邻域内连续, 问

- 1) $\varphi(x, y)$ 应满足什么条件才能使 $f_x(0, 0)$ 和 $f_y(0, 0)$ 都存在?
- 2) 在上述条件下 $f(x, y)$ 在 $(0, 0)$ 点是否可微?

【解】1) 由于 $\lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{|\Delta x| \varphi(\Delta x, 0)}{\Delta x} = \begin{cases} \varphi(0, 0), & \text{当 } \Delta x \rightarrow 0^+, \\ -\varphi(0, 0), & \text{当 } \Delta x \rightarrow 0^-, \end{cases}$

由此可知, 当 $\varphi(0, 0) = 0$ 时, $f_x(0, 0)$ 和 $f_y(0, 0)$ 都存在, 且为零.

2) 当 $\varphi(0, 0) = 0$ 时,

$$\begin{aligned} & \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{[f(\Delta x, \Delta y) - f(0, 0)] - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y]}{(\Delta x)^2 + (\Delta y)^2} \\ &= \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{|\Delta x - \Delta y| \varphi(\Delta x, \Delta y)}{(\Delta x)^2 + (\Delta y)^2} = 0 \end{aligned}$$

$$\leq \frac{|\Delta x|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} + \frac{|\Delta y|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \leq 1 + 1$$

【例7】设 $f_x(x_0, y_0)$ 存在, $f_y(x, y)$ 在点 (x_0, y_0) 处连续, 证明

$f(x, y)$ 在点 (x_0, y_0) 处可微.

【分析】只要证 $\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + o(\rho)$

【证】 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

$$= f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) + f(x_0 + \Delta x, y_0) - f(x_0, y_0)$$

由拉格朗日中值定理得

$$① f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) = f'_y(x_0 + \Delta x, y_0 + \theta\Delta y)\Delta y$$

由 $f'_x(x_0, y_0)$ 存在可知

$$② f(x_0 + \Delta x, y_0) - f(x_0, y_0) = f'_x(x_0, y_0)\Delta x + \alpha_2\Delta x$$

则 $\Delta z = f_y(x_0 + \Delta x, y_0 + \theta\Delta y)\Delta y + f_x(x_0, y_0)\Delta x + \alpha_2\Delta x$

$$f_y(x_0, y_0)$$

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$$0 < \theta < 1 = f'(\xi)\Delta x$$

$$f(x_0 + \Delta x) - f(x_0) = f'(x_0 + \theta\Delta x)\Delta x$$

$$\rightarrow f_x(x_0, y_0) + q$$

$$\lim_{x \rightarrow x_0} f(x) = A$$

$$\rightarrow f(x) = A + q$$

又由 $f'_y(x, y)$ 在点 (x_0, y_0) 处连续可知

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \underline{f'_y(x_0 + \Delta x, y_0 + \theta \Delta y)} = \underline{f'_y(x_0, y_0)}$$

$$\underline{f'_y(x_0 + \Delta x, y_0 + \theta \Delta y)} = \underline{f'_y(x_0, y_0)} + \underline{\alpha_1}$$

$$\Delta z = \underline{f_y(x_0, y_0)\Delta y} + \cancel{\alpha_1 \Delta y} + \underline{f_x(x_0, y_0)\Delta x} + \cancel{\alpha_2 \Delta x}$$

$$\left| \frac{\alpha_1 \Delta y + \alpha_2 \Delta x}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \right| \leq \frac{|\alpha_1| |\Delta y| + |\alpha_2| |\Delta x|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \leq \underline{|\alpha_1| + |\alpha_2|} \rightarrow 0.$$

$$\Delta z = \underline{f_x(x_0, y_0)\Delta x} + \underline{f_y(x_0, y_0)\Delta y} + \underline{o(\rho)}$$

$$\frac{\varphi}{x} \rightarrow 0 \quad \varphi = o(x)$$

$$+ o(\rho)$$



还不关注，
你就慢了



祝同学们 考研路上一路顺利!



还不关注，
你就慢了

